Vipul Sapra

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EDUCATION

University of Cincinnati, Carl H. Lindner College of Business — **Cincinnati, OH**
Master of Science in Information Systems and Graduate Certificate in Data Science — **April 2022**
Cumulative GPA: 3.95 / 4.0

Pune University, Maharashtra Institute of Technology — **Pune, India**
Bachelor of Engineering in Petrochemical — **June 2017**

TECHNICAL SKILLS

Concepts: Generative AI & LLMs, Agentic AI, Autonomous Workflows, MLOps, Autonomous Anomaly Detection, Predictive Modeling, Time-Series Forecasting (SARIMA), Data Engineering, ETL/ELT Pipelines, Dimensional Modeling (Star/Snowflake, SCD Type 1/2), Data Governance & Lineage, Automated Testing & QA, Performance Optimization, Semantic Modeling.

Programming Languages & Tools: KIRO IDE, Copilot, Python, PySpark, XGBoost, Scikit-Learn, SARIMA, Azure (ADF, Databricks, Synapse, Azure SQL), AWS (S3, Redshift), Snowflake, Apache Spark, Power BI (DAX), Tableau, Collibra, SQL, Power Automate, Power Apps.

Certifications: Data Science Graduate Certificate – University of Cincinnati

PROFESSIONAL EXPERIENCE

Charter Communications — Charlotte, NC

Senior Data, Application & Automation Engineer - AI & Cloud Platforms — **May 2022 – Present**

- **Architected and deployed custom autonomous agentic workflows** (KIRO IDE, prompt orchestration, Copilot) to automate schema discovery, data profiling, and code migration, reducing manual delivery turnaround by 40-50% while ensuring strict adherence to enterprise release standards.
- **Engineered autonomous machine learning anomaly detection frameworks** identifying schema variations, distribution drift, and ETL discrepancies with 95% precision, directly safeguarding downstream BI, database integrations, and pipeline integrity for critical business applications.
- **Developed high-throughput distributed cloud data pipelines** and dimensional models (Star/Snowflake schemas, SCD Type 1/2) utilizing Azure (ADF, Databricks, Synapse, SQL) and AWS (S3, Redshift) to crunch hundreds of terabytes with 99.9% data availability SLAs.
- **Engineered automated testing and evaluation frameworks** for conversational AI platforms (SpectrumIQ) and workflow automations, cutting manual QA turnaround by 50% and accelerating production release cycles across distributed enterprise environments.
- **Built automated data quality reconciliation firewalls** and integrated Collibra data lineage to reduce operational turnaround by over 80%, ensuring compliance with robust data governance, security, and access control policies.

Data Scientist & Machine Learning Specialist — **May 2022 – Present**

- **Built and deployed predictive machine learning models** (XGBoost, Random Forests, Scikit-Learn) predicting customer behavior, delivering actionable telemetry and data-driven insights that improved overall service reliability by 20%.
- **Architected advanced time-series forecasting models** (SARIMA) to predict truck rolls and field support demand, optimizing workforce capacity allocation, resource scheduling, and reducing operational response times by 20%.
- **Translated complex transactional telemetry into executive strategies** by developing semantic models and Key Operating Metric (KOM) dashboards in Power BI (DAX) and Tableau to support data-driven decision making across stakeholder groups.

Business Intelligence & Decision Analytics Specialist — **May 2022 – Present**

- **Designed and optimized complex SQL queries, stored procedures** and data transformations across relational database systems to support robust business applications, reporting products, and automated integration workflows.
- **Created interactive executive dashboards and semantic data models** using Power BI and Tableau, translating intricate operational metrics into actionable insights and strategic visibility for leadership.
- **Streamlined data-ingestion, extraction, transformation, and loading** (ETL) processes, applying rigorous data quality validation checks and performance tuning to eliminate recurring defects and enhance system reliability.
- **Collaborated with cross-functional technical teams** to troubleshoot complex defects, perform root-cause analysis, and deliver maintainable enhancements for cloud-based reporting and automation platforms.

Ameritas Insurance — Cincinnati, OH

Data Governance Intern — **September 2021 – April 2022**

- **Co-engineered the enterprise-wide Data Advocacy & Governance Program** across insurance portfolios, establishing foundational frameworks for data democratization and organizational accountability.

- **Architected a comprehensive Data Intelligence ecosystem in Collibra**, establishing a centralized Data Dictionary, automated Data Quality rules, and end-to-end **Data Lineage** to ensure regulatory compliance and transparency for financial and insurance assets.

Accenture **Pune, India**

Test Engineering Analyst **September 2017 – December 2020**

- **Orchestrated end-to-end Quality Assurance and system integration testing** for mission-critical client applications at **Allianz Insurance**, leveraging Agile SDLC methodologies to ensure rapid, zero-defect feature deployments.
- **Spearheaded the Health Insurance module QA strategy**, directing cross-functional efforts for regression testing, test data validation, and the seamless rollout of complex new business functionalities.
- **Drove operational excellence by automating core regression test suites**, significantly increasing engineering productivity and reducing manual testing overhead through scalable automation frameworks.